

MODULE 04

Rheumatologic Arthropathies: A Pattern Recognition Approach

RA · Psoriatic Arthritis · Gout · CPPD · Ankylosing Spondylitis · Sacroiliitis

1.0 AMA PRA Cat 1

General MSK

CME Article Review

LEARNING OBJECTIVES

1. Apply a six-axis framework (distribution, bone density, erosion pattern, joint space, new bone formation, soft tissue) to any arthropathy radiograph.
2. Identify distinguishing radiographic features of rheumatoid arthritis, psoriatic arthritis, gout, CPPD, and ankylosing spondylitis.
3. Recognize joint-specific patterns: pancarpal RA, pencil-in-cup PSA, overhanging-edge gout erosions, and CPPD chondrocalcinosis.
4. Differentiate symmetric (AS, IBD-associated) from asymmetric (PsA, reactive arthritis) sacroiliitis patterns.
5. Apply the modified New York criteria to grade sacroiliitis (Grade 0–4).
6. Recognize hallmark hand deformities of RA and the clinical implications of atlantoaxial subluxation.

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1. A Six-Axis Framework for Arthropathy Analysis

Rather than attempting to recognize each arthropathy as a memorized gestalt, a systematic six-axis approach allows reliable differentiation even in atypical or overlapping cases. For every arthritis study, deliberately assess: (1) distribution pattern, (2) periarticular bone density, (3) erosion morphology, (4) joint space change, (5) new bone formation, and (6) soft tissue/calcification pattern. Most diagnostic confusion in arthropathy imaging comes from skipping one of these axes — for example, failing to note preserved bone density (which argues strongly against RA) or missing a subtle overhanging erosion edge (pathognomonic for gout).

The Six-Axis Framework

Axis	Key Question	Discriminating Diagnoses
1. Distribution	Symmetric vs. asymmetric? Ray pattern? Axial involvement?	Symmetric → RA, CPPD; Asymmetric → PsA, gout; Ray pattern → PsA
2. Bone density	Periarticular osteopenia present, or preserved?	Osteopenia → RA, JIA; Preserved → gout, PsA, OA, CPPD
3. Erosion pattern	Marginal? Central ("seagull")? Overhanging edge?	Marginal → RA; Central → erosive OA; Overhanging → gout
4. Joint space	Uniform narrowing? Focal? Pan-compartmental?	Uniform → RA; Focal → OA; Pancarpal uniform → RA
5. New bone formation	Osteophytes? Syndesmophytes? Periostitis?	Osteophytes → OA; Syndesmophytes → AS; Periostitis → PsA, reactive arthritis
6. Soft tissue / calcification	Tophi present? Calcification pattern?	Tophi → gout; Chondrocalcinosis → CPPD; Calcinosis cutis → scleroderma

- Working through all six axes explicitly — even when the diagnosis seems obvious at first glance — catches the cases where two features point in different directions (e.g. erosive change with preserved bone density, which should immediately raise PsA or gout over RA).

2. Rheumatoid Arthritis & Psoriatic Arthritis — Key Distinguishing Features

RA and PsA can both produce hand and wrist erosive arthropathy, but the distribution, bone density, and erosion morphology differ in ways that are usually reliable even on a single radiograph. RA is the prototypical symmetric, MCP/PIP/wrist-predominant erosive arthropathy with periarticular osteopenia and no new bone formation. PsA, by contrast, classically spares bone density, involves the DIP joints, follows a ray (whole-digit) distribution, and produces periostitis rather than pure erosion.



Rheumatoid arthritis: symmetric MCP and pancarpal involvement with periarticular osteopenia and marginal erosions, DIP joints spared.

Rheumatoid Arthritis vs. Psoriatic Arthritis

Feature	Rheumatoid Arthritis	Psoriatic Arthritis
Distribution	Symmetric; MCP/PIP/wrist predominant; DIP spared	Asymmetric; DIP involved; ray (whole-digit) pattern
Bone density	Periarticular osteopenia (hallmark)	Preserved — key differentiating feature
Erosion	Marginal, at the "bare area"; radial MCP heads first	Pencil-in-cup at DIP; "mouse ear" pattern; can be central
Joint space	Pancarpal collapse; uniform narrowing	Preserved, or eccentric; DIP may undergo ankylosis
New bone	Absent — no osteophytes or periostitis	Periostitis; enthesophytes; fluffy periosteal new bone
SI joints	Not involved	Asymmetric sacroiliitis
Spine	Step-ladder subaxial subluxation; atlantoaxial subluxation	Bulky, non-marginal, asymmetric paravertebral ossification

Hallmark RA Hand Deformities

Long-standing RA produces a recognizable set of hand deformities from chronic synovitis, ligamentous laxity, and tendon imbalance. These are typically clinical/radiographic correlates rather than acute findings, but recognizing them on radiographs supports a chronic RA diagnosis and helps date disease severity.



Boutonniere deformity: PIP joint flexion with compensatory DIP hyperextension, from central slip extensor tendon disruption.

! Preserved periarticular bone density in an erosive hand arthropathy should immediately shift the differential away from RA and toward PsA, gout, or erosive osteoarthritis — density is one of the most reliable single discriminators across the inflammatory arthropathies.

Key References

- Resnick D, Niwayama G. Diagnosis of Bone and Joint Disorders. 4th ed. Saunders 2002.
- Sudo ■ -Szopi ■ ska I, Kwiatkowska B, Prochorec-Sobieszek M. Imaging of psoriatic arthritis. Pol J Radiol 2013.

3. Crystal Deposition Arthropathy — Gout vs. CPPD vs. HADD

The crystal deposition arthropathies — gout (monosodium urate), CPPD (calcium pyrophosphate), and HADD (calcium hydroxyapatite) — each have a distinct radiographic signature based on where the crystal deposits and how the joint responds. Gout characteristically preserves the joint space until very late disease and produces eccentric, sclerotic-margined "overhanging edge" erosions at a distance from the joint line. CPPD deposits within fibrocartilage and hyaline cartilage, producing chondrocalcinosis, and favors atypical joint distributions (radiocarpal, patellofemoral) compared to primary osteoarthritis.



Gouty arthropathy of the knee: periarticular soft tissue tophus with preserved joint space.

Crystal Arthropathy — Rapid Comparison

Feature	CPPD	HADD (Calcium Apatite)	Gout (Urate)
Crystal	Calcium pyrophosphate	Calcium hydroxyapatite	Monosodium urate
Radiographic calcification	Fibrocartilage/hyaline cartilage; chondrocalcinosis	Peritendinous/peribursal; amorphous "chalk-dust"	Not directly visible (tophi are soft-tissue density unless secondarily calcified)
Key location	TFC (wrist), menisci (knee), pubic symphysis	Rotator cuff (supraspinatus), hip, gluteus minimus	1st MTP, ankle, knee, olecranon, wrist
Erosion	Mild; SLAC-pattern collapse at wrist	None typically; acute calcific deposits may rarely erode	Overhanging edge with sclerotic margin
Joint space	Narrowed; classically patellofemoral-predominant at the knee	Preserved	Preserved until very late disease



CPPD of the knee: uniform tricompartmental joint space narrowing, a distribution distinct from the medial-predominant pattern typical of primary osteoarthritis.

- An overhanging edge erosion with a sclerotic margin and preserved joint space is pathognomonic for gout. Patellofemoral-predominant or uniform tricompartmental knee narrowing (rather than medial-predominant) should raise CPPD over primary osteoarthritis.

Key References

- Steinbach LS. Calcium pyrophosphate dihydrate and calcium hydroxyapatite crystal deposition diseases: imaging perspectives. Radiol Clin North Am 2004.
- Dalbeth N, Doyle AJ. Imaging tools to measure treatment response in gout. Rheumatology 2018.

4. Sacroiliitis Patterns & Ankylosing Spondylitis

The single most useful discriminator among the spondyloarthropathies is the symmetry of sacroiliitis: ankylosing spondylitis and inflammatory bowel disease-associated arthropathy produce bilateral, symmetric sacroiliitis progressing through erosion, sclerosis, and eventual complete ankylosis, while psoriatic arthritis and reactive arthritis characteristically produce asymmetric or unilateral involvement. Spinal involvement follows the same symmetry pattern: AS produces delicate, marginal, vertical syndesmophytes ("bamboo spine"), while PsA and reactive arthritis produce bulky, non-marginal, asymmetric paravertebral ossification.



Asymmetric/unilateral sacroiliitis, a pattern favoring psoriatic or reactive arthritis over ankylosing spondylitis.

Modified New York Criteria — Sacroiliitis Grading

Grade	Findings
0	Normal
1	Suspicious changes (equivocal)
2	Minimal sclerosis or small erosions, without joint space alteration
3	Definite erosion, sclerosis, widening, narrowing, or partial ankylosis
4	Total ankylosis

! A triangular, iliac-sided sclerosis at the SI joint without erosion or joint space change, particularly in a childbearing-age woman, suggests osteitis condensans ilii — a benign mechanical finding, not an inflammatory sacroiliitis — and should not be mistaken for AS.

Key References

- van der Linden S, Valkenburg HA, Cats A. Evaluation of diagnostic criteria for ankylosing spondylitis: a proposal for modification of the New York criteria. *Arthritis Rheum* 1984.
- Braun J, Sieper J. Ankylosing spondylitis. *Lancet* 2007.

5. RA of the Cervical Spine — Atlantoaxial Subluxation

Chronic RA pannus formation around the odontoid and transverse ligament of C1 can erode the ligament, allowing anterior subluxation of C1 on C2. This is assessed on a lateral cervical spine radiograph by measuring the atlantodental interval (ADI) — the distance between the posterior margin of the anterior arch of C1 and the anterior margin of the odontoid process. This measurement has direct implications for patient safety, particularly around anesthesia and intubation, where neck manipulation in an unrecognized unstable atlantoaxial joint can cause cervical cord injury.



Lateral cervical spine radiograph used to measure the atlantodental interval (ADI) for evaluation of atlantoaxial subluxation in RA.

Atlantodental Interval — Clinical Thresholds

ADI	Significance
≤ 3 mm (adult)	Normal
> 3 mm (adult)	Atlantoaxial subluxation — transverse ligament compromise; neurosurgical referral indicated
> 9–10 mm	High risk of spinal cord compromise; surgical stabilization typically required

! ADI widening in RA is a referral and surgical-planning finding, not necessarily an emergent one — clinical correlation for neurological signs determines urgency. Always measure and report the ADI explicitly on any cervical spine study in a patient with known RA, even when not the primary indication for the exam.

RA of the Hip — Protrusio Acetabuli

In addition to the cervical spine, chronic RA can produce protrusio acetabuli at the hip — medial migration of the femoral head beyond Kohler's line, from progressive acetabular bone softening and remodeling under chronic synovitis.



Bilateral protrusio acetabuli in chronic rheumatoid arthritis, with medial migration of the femoral heads beyond Kohler's line.

Key References

- Bouchaud-Chabot A, Lioté F. Cervical spine involvement in rheumatoid arthritis. *Joint Bone Spine* 2002.
- Rana AQ, Dejaegher J, Verschueren P. Atlantoaxial subluxation in rheumatoid arthritis: clinical and radiological considerations. *Acta Neurochir* 2015.